RetailTech Analytics Challenge

Jinyang Si | Data Engineering Class

Executive Summary

Analyzed 25,000+ Amazon reviews across five product categories. Built classification systems to automate:

- Product categorization
- Rating prediction
- Custom logistic modeling
- Developed ROI-backed roadmap for implementation. Expected 3-year ROI: 172%, break-even in ~8 months

Challenge 1: Product Categorization

Ensemble

Accuracy: 85.1%

Key Insight

- Misclassifications often involve Home & Kitchen overlapping with other categories
- PCA reveals category overlap explains confusion

Recommendation

- Fine-tune embeddings using domain-specific data
- Implement human-in-the-loop review for borderline cases

Challenge 2: Rating Prediction

SVR with engineered features

Accuracy: 81.1% within ±1 star

Key Features

- Sentiment scores (VADER)
- Review length, helpfulness votes

Recommendation

- Use this model to flag reviews with mismatched tone & rating
- Improve review authenticity & product messaging

Challenge 4:
Logistic Regression from Scratch

Capabilities

- Binary + Multi-class (OvR & Softmax)
- L1/L2 regularization
- Convergence visualized with loss curves

Insight

- Custom model reached ~64% accuracy vs. 85% from scikit-learn
- Great for learning, but not yet production-ready

Recommendation

- Use custom model for R&D, not deployment
- Extend with momentum-based optimizers

Challenge 5: ROI & Strategy

Analysis

Implementation Cost: $73K annually
Annual Benefit: $115K
Break-even: ~8 months
3-Year ROI: 172%

Strategy Wins

- Reduced manual review workload
- Increased customer retention
- More effective product onboarding

Final Recommendations

01

Operational

- Automate review tagging → boost content team speed
- Sentiment-check rating mismatches → improve trust
- Review dashboard → live insights on product & customer health

02

Customer Segmentation

- Persona 1: “Detail Seekers” – Long reviews, emotionally nuanced, value quality
- Persona 2: “One-liners” – Concise, sentiment-rich, price-sensitive
- Persona 3: “Balanced Critics” – Moderate tone, helpful voters, loyal shoppers

03

Tactics

- Tailored email marketing
- Product bundling by sentiment tone
- A/B test copy for different personas

Next Steps

Assesment	Planning	Implementation	Monitoring
Finalize model deployment	Connect systems to backend product database	Launch internal dashboard	Conduct A/B testing for sentiment-aligned messaging